

Use Case Number	LNA01	
Use Case Name	Login or Create New Account 01	
Actor	Patients, Nurses & Physicians	
Goal (1 phrase)	Nurse staff to register new patients or manage previous patients' profile accounts.	
Overview and scope	The system uses an authentication/authorization system when users log in to access patient charts. An assigned user ID is required for access, and multi-factor authentication is used for remote access.	
Level	Primary	
Preconditions	Office personnel are logged into the system.	
Postconditions in words (write in passive and past tense)	1. User has been assigned a login ID 2. Remote users have the multi-factor authentication app installed. 3. New patient chart is created. 4.	
Trigger	A new or returning patient arrives at the clinic.	
Included Use Cases	None	
Extending Use Cases	Schedule appointment Process payment	
MAIN SUCCESSFUL SCENARIO for this Use Case in numbered sequence Reference included use cases in this section using: INCLUDE <i>ius_name</i> *	Actor Action	System Action
	1. A patient arrives at the clinic.	2. Office staff member logs into the system using their assigned user ID
	3. The office staff member selects the option to register a new patient.	4. System brings up a new account page
	5. Office staff enters the patient's information, including name, date of birth, address, insurance details, emergency contact, and phone number.	6. The system validates the input data and ensures no duplicate.
	7. Actor confirms information	8. System logs creation event in
OTHER SUCCESSFUL SCENARIOS (Specify any successful variations of the normal	Step	Branching Action
	2a A previous patient comes in to the clinic	Office staff member logs into the system using their assigned user ID

execution path, including any extension points using EXTEND <i>eus_name</i>)	2b The office staff member selects the option to log into patient history	System brings up a patient's account.
	2c Staff enters reasons for visit	9a. System inputs data
	8b Customer pays by check	9b. System indicates check and enter amount
UNSUCCESSFUL SCENARIOS (Erroneous situations)	Conditions	Actions
	*a Staff enters password without MFA login	System denies log in entry
	B. Staff tries to create new account for existing patient	System denies duplicate data input
	9c.Patient leaves unexpectedly, staff cancel entries of new account	System aborts the process
Priority in scheduling	Primary	
Frequency	A patient comes to the clinic and needs access	
Business rules and data logic	- Each user (nurse, physician, office staff) must be assigned a unique user ID. - Every staff member must authenticate using MFA provided via a secure app. - Each new patient must be assigned a unique medical number automatically by the system.	
Other non-functional requirements	- All personal information given will be encrypted in the system and only allows access to necessary staff. - Email, password and past visits can also be saved for easy log in in future visits. - Passwords will need to be updated at least every six months.	
Superordinates	N/A	
Developer	Tobi Bakare	
Creation & last modified date	05.14.2024 05.16.2024	
Other Comments	N/A	

Use Case Number	SAP02
Use Case Name	Schedule a new appointment 02

Actor	Patients, Nurses & Physicians	
Goal (1 phrase)	Nurse needs to schedule a new appointment or update an existing one.	
Overview and scope	This use case outlines the process for securely accessing the system, managing patient charts, and scheduling or updating appointments through an integrated calendar interface. The system ensures secure access through an authentication and authorization process and allows office staff to handle patient registrations and manage appointments efficiently.	
Level	Primary	
Preconditions	- Office personnel are logged into the system. - Users have been assigned a login ID. - Office personnel have installed the multi-factor authentication app.	
Postconditions in words (write in <i>passive and present tense</i>)	- Appointment details are updated in the system's calendar. - Nurse gains access to the system after successfully completing the authentication process. - New patient appointment is created, and the patient is assigned a unique medical number.	
Trigger	Need to schedule a new appointment or update an existing one.	
Included Use Cases	Log in to account	
Extending Use Cases	Process Payment	
MAIN SUCCESSFUL SCENARIO <i>for this Use Case</i> in numbered sequence Reference included use cases in this section using: INCLUDE <i>ius_name</i> *	Actor Action	System Action
	1. Office staff need to schedule a new appointment or update an existing one.	2. The system validates the input
	3. Office staff accesses the system's calendar interface.	4. System shows the available dates
	5. Office staff selects the desired time slot and enters the appointment details	6. The system validates the changes, fills dates and updates the appointment details.
	6. Staff confirms details	7. System updates the calendar and logs the update event.
	8.	

OTHER SUCCESSFUL SCENARIOS (Specify any successful variations of the normal execution path, including any extension points using EXTEND <i>eus_name</i>)	Step	Branching Action
	2a Office staff searches for the existing appointment in the calendar.	System shows appointments
	2b Office staff selects the appointment and makes necessary changes	The system validates the changes and updates the appointment details.
		System updates the calendar and logs the update event
UNSUCCESSFUL SCENARIOS (Erroneous situations)	Conditions	Actions
	*a Staff attempts to enter appointment on already taken time slot	System denies duplication appointment on same time and date
	8c. Office staff input appointment on clinic holiday	System flags holiday
	Office staff tries to create two appointments for the same patient with the same physician	System aborts duplicate appointment for same patient
Priority in scheduling	Primary	
Frequency	Every time a patient needs an appointment or nurse needs to access an existing one.	
Business rules and data logic	- Each user (nurse, physician, office staff) must be assigned a unique user ID. - Every staff member must authenticate using MFA provided via a secure app. - Each new patient must be assigned a unique medical number automatically by the system - All appointments updates are reflected in real-time on the system's calendar. - All appointment details are checked to avoid conflicts.	
Other non-functional requirements	- All appointments info given will be encrypted in the system and only allow access to necessary staff. - Email, password and past visits can also be saved for easy log in in future visits. - Passwords will need to be updated at least every six months.	
Superordinates	N/A	
Developer	Tobi Bakare	

Creation & last modified date	05.14.2024 05.16.2024
Other Comments	N/A

Use Case Number	PP03	
Use Case Name	Process Payment 03	
Actor	Patients & Nurses	
Goal (1 phrase)	Make payment for patient visits to the clinic.	
Overview and scope	After a patient visit, the office staff processes payments and updates the system accordingly. The system calculates costs, verifies insurance coverage, and produces receipts for transactions.	
Level	Extended	
Preconditions	- Office personnel are logged into the system. - Patient visit information is recorded in the system.	
Postconditions in words (written <i>passive and past tense</i>)	- Patient payment chart is updated - Payment is processed - Transaction details are updated in the system.	
Trigger	Patient completes a visit and provides payment.	
Included Use Cases	Login Account Set a new appointment	
Extending Use Cases	N/A	
MAIN SUCCESSFUL SCENARIO <i>for this Use Case</i> in numbered sequence Reference included use cases in this section using: INCLUDE <i>ius_name</i> *	Actor Action	System Action
	1. Patient finishes up a visit and proceeds to make a payment.	2. System brings up patient invoice data
	3. Office staff access the payment processing module in the system.	4. System calculates the total cost, considering services provided and insurance coverage.
	5. Office staff verifies the calculated amount with the patient	6. System confirms the insurance details.

	6. Patient provides payment through available methods	7. System selects payment method
	8. Office staff processes the payment	9. The system updates the transaction details.
	10. Office staff ask patient if they want a receipt or not	11. The system generates a receipt for the patient
	12. Office staff provide the receipt to the patient.	
OTHER SUCCESSFUL SCENARIOS (Specify any successful variations of the normal execution path, including any extension points using EXTEND <i>eus_name</i>)	Step	Branching Action
	2a Patient set up payment plan form the options	Systems sets it up with reminders for patient
	8a Patient pays by cash	9a. System indicates cash and enter amount
	8b Patient pays by check	9b. System indicates check and enter amount
UNSUCCESSFUL SCENARIOS (Erroneous situations)	Conditions	Actions
	*a Patient's insurance is expired or has issues	System alerts the staff of issues
	8c. Credit card fails verification	System shows other payment options
	9c.No other methods works for patient	System aborts payment and alerts office staff for further assistance.
Priority in scheduling	Secondary	
Frequency	Every time a bill is due for payment	
Business rules and data logic	1. Each user (nurse, physician, office staff) must be assigned a unique user ID. 2. Every staff member must authenticate using MFA provided via a secure app. 3. Each new patient must be assigned a unique medical number automatically by the system	
Other non-functional requirements	1. Encrypt sensitive payment data throughout process 2. Systems provides a user -friendly interface with various payment options for all pay groups. 3. Ensure the system complies with relevant healthcare regulations	
Superordinates	N/A	

Developer	Tobi Bakare
Creation & last modified date	05.14.2024 05.16.2024
Other Comments	N/A